

Microcontroller-based Bit Error Rate Test set

Vishnu Barath

3rd Year Project Interim Report

Department of Electronic &
Electrical Engineering

UCL

Supervisor: Dr Martyn Fice

November 28, 2025

I have read and understood UCL's and the Department's statements and guidelines concerning plagiarism.

I declare that all material described in this report is my own work except where explicitly and individually indicated in the text. This includes ideas described in the text, figures and computer programs.

This report contains _ pages (excluding this page and the appendices) and _ words.

Signed:

Contents

1	Project Description	2
2	Literature Review	2

1 Project Description

Channels for digital communication as susceptible to incorrect data transmission arising due many possible factors such as noise or interference.[1] Communication networks are continuously expanding in capacity and speed and being able to ensure data integrity is paramount. The most common example of such a system is the internet, which uses undersea optical fibres for data transmission. This project aims to construct a system that can

2 Literature Review

References

- [1] S. M. J. Alam, M. R. Alam, G. Hu, and M. Z. Mehrab, “Bit error rate optimization in fiber optic communications,” *International Journal of Machine Learning and Computing*, vol. 1, no. 5, pp. 435–440, 2011, doi:<http://dx.doi.org/10.7763/IJMLC.2011.V1.65>.